$$\alpha_t = 1 - \beta_t, \quad \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$$

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

$$L_\epsilon = \mathbb{E} \| \epsilon - \epsilon_\theta(z_t, t, y) \|_2^2$$